Equipment	No. of breakages
Burette	45
Conical flask	75
Test tube	150
Pipette	30

Draw Pareto Diagram for the above data. Which equipment requires more attention in order to reduce breakages?

Solution:

Since we have to find the vital few among the several, we draw Pareto diagram.

Step 1 : Arrange the equipment according to the descending order of the number of breakages.

Step 2 : Find the percentage of breakages for each equipment using the formula $= \frac{No.\ of\ Breakages}{Total\ No.\ of\ Breakages} \times 100$

Step 3 : Calculate cumulative percentage for each equipment.

Step 4 : Mark the equipment along the X-axis and the number of breakages along the Y-axis. Construct an attached simple bar diagram to this data. In an attached simple bar diagram, the vertical bars are erected adjacently.

Step 5 : Mark the cumulative no. of breakages for each equipment corresponding to the mid-point of the respective vertical bar.

Step 6 : Draw a free hand curve joining those plotted points.

Equipment	No. of Breakages (Frequency)	No. of Breakages in percentage	Cumulative No. of Breakages in percentage
Test tube	150	$\frac{150}{300} \times 100 = 50$	50
Conical flask	75	$\frac{75}{300} \times 100 = 25$	75
Burette	45	$\frac{45}{300} \times 100 = 15$	90
Pipette	30	$\frac{30}{300} \times 100 = 10$	100
Total	**300**	**100**	